

Literature-Implied Status: Warped Cone Coarse Baum–Connes

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Status

This is a literature-implied partial answer packet, not an original proof from the run. Druţu–Nowak, arXiv:1501.03473, conjectured that the coarse assembly map is not surjective for warped cones arising from their spectral-gap ghost-projection construction. Kitsios–Schick–Vigolo, arXiv:2504.21811, explicitly identify the Roe/Druţu–Nowak conjecture and prove the unified warped-cone non-surjectivity statement under the extra hypotheses that the acting group has property A and the action is free and strongly ergodic.

This does not settle the full 2015 conjecture, but it gives a direct literature answer for an important natural subcase, including the standard free-subgroup action on $SU(2, \mathbb{C})$ highlighted in the supporting paper.

Original Problem

The original source is:

C. Druţu and P. W. Nowak, *Kazhdan projections, random walks and ergodic theorems*, arXiv:1501.03473.

After proving that spectral-gap actions yield non-compact ghost projections for warped cones, the paper states Conjecture 6.7: under the same hypotheses, the coarse assembly map should not be surjective for the warped cone $X = \mathcal{O}_G M$.

class represented by $\mathcal{O} \otimes p$ in $K_*^*(\mathcal{O} \rtimes G)$, where p is a rank one projection, is not in the image of the coarse assembly map. At present such expanders provide the only known bounded geometry counterexamples to the coarse Baum–Connes conjecture.

Let G, M satisfy the assumptions of Theorem 6.6.

Conjecture 6.7. *The coarse assembly map (17) is not surjective for the warped cone $X = \mathcal{O}_G M$.*

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Supporting Theorem

The supporting source is:

C. Kitsios, T. Schick, and F. Vigolo, *Coarse Baum–Connes and warped cones: failure of surjectivity in odd degree*, arXiv:2504.21811.

The 2025 paper formulates the relevant conjecture as Roe/Druţu–Nowak Conjecture 7.7 for a compact Riemannian manifold M and a measure-preserving action by diffeomorphisms with spectral gap.

well-defined.

The warped cone $\mathcal{O}_\Gamma M$ has a rather nice local geometry. For instance, if the action is free and by isometries, one can show that the ball of radius R centered at a point (x, t) will converge (in the Gromov–Hausdorff sense) to the ball of radius R in $\mathbb{R}^{\dim(M)} \times \Gamma$ as $t \rightarrow \infty$ [8, 40]. On the other hand, the global geometric features of $\mathcal{O}_\Gamma M$ are very sensitive to the dynamical properties of the action [12, 28, 33, 40]. In particular, Roe claimed already in [33] that warped cones could be used to construct counterexamples to the coarse Baum–Connes conjecture—this was one of Roe’s primary motivations for introducing this construction. This conjecture was later made precise by Druţu and Nowak as follows:

Conjecture A (Roe, Druţu–Nowak [10, Conjecture 7.7]). *Let M be a compact Riemannian manifold and $\Gamma \curvearrowright M$ a measure-preserving action by diffeomorphisms. If the action has a spectral gap, then μ_c is not surjective for $\mathcal{O}_\Gamma M$.*

If this conjecture was true, this would be a very interesting source of examples indeed. For instance, if $\Gamma = \pi_1(N)$ for some compact manifold N , then $\mathcal{O}_\Gamma M$ would be coarsely equivalent to a complete Riemannian manifold with compact boundary (to see this, it is enough to observe that $\mathcal{O}_\Gamma M$ is coarsely equivalent to a “foliated warped cone” [33, Lemma 1.12]). Replacing the boundary with a cusp would also yield a complete Riemannian manifold without boundary and with compact core, which by a Mayer–Vietoris argument would still violate the coarse Baum–Connes conjecture. This setup is quite different from the previous source of counterexamples.

Examples of actions as above are easily obtained by appropriately taking two cocompact lattices Γ, Λ in some higher rank Lie group G and let Γ act on $M := G/\Lambda$.

¹Note that we only stated the coarse Baum–Connes conjecture for spaces of bounded geometry. For spaces of unbounded geometry a few other obstructions are known.

Its Theorem C says that if Γ has property A and $\Gamma \curvearrowright M$ is a free strongly ergodic action by diffeomorphisms, then

$$\mu_c : KX_1(\mathcal{O}_\Gamma M) \longrightarrow K_1(C_{\text{Roe}}^*(\mathcal{O}_\Gamma M))$$

is not surjective. The authors note that this does not exactly match the Druţu–Nowak conjecture: it adds property A and freeness, while replacing spectral gap by the weaker strong ergodicity condition. They also note that their $\text{SU}(2, \mathbb{C})$ action satisfies the hypotheses.

An even simpler example is to take Γ to be a non-abelian free subgroup of $\mathrm{SU}(2, \overline{\mathbb{Q}})$ and let $\Gamma \curvearrowright \mathrm{SU}(2, \mathbb{C})$ be the action by left multiplication. This free, isometric action has spectral gap by deep work of Bourgain–Gamburd [4].

The first result of this paper appears to provide evidence against [Conjecture A](#). Namely, the motivation behind the claim of Druţu–Nowak (and presumably also Roe’s) is that under those assumptions the Roe algebra $C_{\mathrm{Roe}}^*(\mathcal{O}_\Gamma M)$ can be shown to contain non-compact ghost projections. We will however show that these projections vanish in K-theory:

Theorem B ([Theorem B](#)). *If $\Gamma \curvearrowright M$ is ergodic and $\mathfrak{G} \in C_{\mathrm{Roe}}^*(\mathcal{O}_\Gamma M)$ is a generalized Druţu–Nowak projection, then $[\mathfrak{G}] = 0$ in $K_0(C_{\mathrm{Roe}}^*(\mathcal{O}_\Gamma M))$.*

We prove [Theorem B](#) combining the naturality properties of K-theory with a change of perspective coming from [9, 23].

As it turns out, the situation is however not at all as dire as it may appear. In fact, the vanishing in K-theory of the Druţu–Nowak projection can be explained via a Mayer–Vietoris argument, which also shows the path to follow to save [Conjecture A](#). In turn, we can then confirm the conjecture of Roe by proving the following.

Theorem C. *Let Γ be a group with property A and $\Gamma \curvearrowright M$ a free and strongly ergodic action by diffeomorphisms. Then the coarse Baum–Connes map*

$$\mu_c: KX_1(\mathcal{O}_\Gamma M) \rightarrow K_1(C_{\mathrm{Roe}}^*(\mathcal{O}_\Gamma M))$$

is not surjective.

Remark. [Theorem C](#) does not entirely align with [Conjecture A](#): on the one hand we requires a few extra conditions (property A, freeness), on the other hand we do not require the full power of spectral gap (strongly ergodic actions suffice). Note that the action $\Gamma \curvearrowright \mathrm{SU}(2, \mathbb{C})$ described above satisfies the hypotheses of [Theorem C](#).

Remark. The failure of the coarse Baum–Connes map being an isomorphism in

Idea of the Supporting Proof

The original Druţu–Nowak route expected the non-compact ghost projection itself to supply an even-degree K-theory obstruction. Kitsios–Schick–Vigolo show that generalized Druţu–Nowak projections actually vanish in K_0 for ergodic actions. They then use a Mayer–Vietoris decomposition of the unified warped cone and sparse-level ghost-projection obstructions to produce an odd-degree class in the Roe algebra that cannot lie in the image of the coarse assembly map.

Implication for arXiv:1501.03473

For the property-A/free/strongly ergodic subcase, Kitsios–Schick–Vigolo give the non-surjectivity conclusion sought by Druţu–Nowak. In particular, spectral gap implies strong ergodicity for measure-preserving actions, and free groups have property A, so the free-subgroup $\mathrm{SU}(2)$ examples fall within this supporting theorem.

Scope and Limitations

This packet should be read as a literature-implied partial answer. The full statement of Conjecture 6.7 in arXiv:1501.03473 is broader: it does not impose property A or freeness. The 2025 theorem

is therefore not a full `literature_already_answered` resolution of the 2015 conjecture.

The scan hit quoting Willett–Yu Problem 5.4 is a same-paper ask-and-answer passage: Druţu–Nowak answer that older request by constructing warped-cone ghost projections. A nearby “coarsely contain expanders” question in the source TeX is inside a discarded `\comment . . . \endcomment` block and was not treated as a published target.

References

- C. Druţu and P. W. Nowak, *Kazhdan projections, random walks and ergodic theorems*, arXiv:1501.03473.
- C. Kitsios, T. Schick, and F. Vigolo, *Coarse Baum-Connes and warped cones: failure of surjectivity in odd degree*, arXiv:2504.21811.
- K. Li, F. Vigolo, and J. Zhang, *A Markovian and Roe-algebraic approach to asymptotic expansion in measure*, arXiv:2008.12572, used only as background for the sparse/integral warped-cone literature trail.

Review Recommendation

Check the status label. The packet records a real later-literature confirmation for a major subcase and for the canonical $SU(2)$ example, while preserving that the unrestricted spectral-gap conjecture is not fully settled here.